Safe and Scalable Web Agent Learning via Recreated Websites

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Paper version: arXiv 2603.10505v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Real-world websites are unsafe to explore, hard to reset, and rarely provide verifiable feedback.*

VeriEnv clones a site, exposes a **Python SDK**, and scores runs from database state, not an LLM judge. Training then happens only in the clone.

Motivation: three obstacles

1. **Unsafe or blocked.** Real sites can be harmed or hit CAPTCHAs.
2. **Loose tasks.** Several valid answers.
3. **Model judges.** Error-prone rewards.

Evaluation setup

WebArena-Lite · 5 sites | M2W-Online · 220 tasks | Qwen3-4B base | LLaMA-3.2-3B-Instruct | GPT-5.2 builds sites | 149 sites · 7,400 tasks | 97 sites for training | rejection fine-tuning

- **All online** success; no offline step accuracy.
- **Held out:** sites overlapping the test splits are excluded from cloning and training.

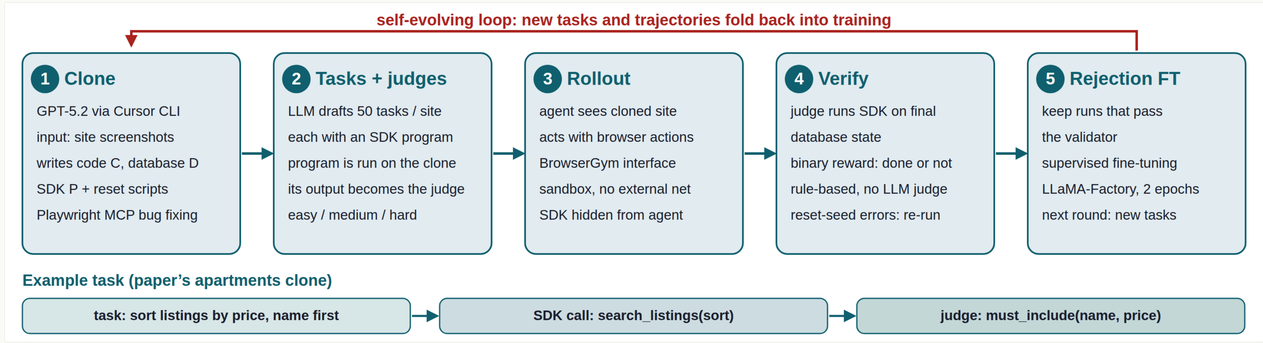
Our read (CUA Radar)

74	★★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; held-out sites; open baselines; cost; validator check. **Not reported:** seeds, budget match, ablation.
- **Flags:** AP2, synthetic environment without a real anchor; code “upon acceptance”.
- Pattern K13. Authors’ numbers, not rerun.

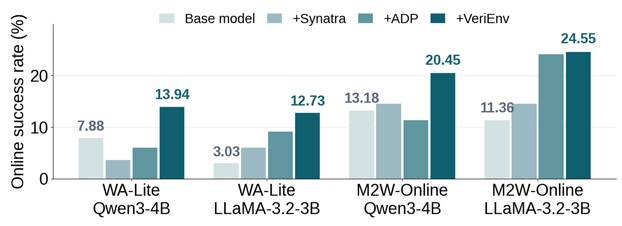
Method: clone the site, verify the task, train on passing runs

- **Sees / acts:** browser observations and actions via BrowserGym; the SDK is never visible to the agent while it runs.
- **Build:** coding agent (GPT-5.2, Cursor CLI) writes code C, database D, SDK P from screenshots; it then tests the site with Playwright MCP and patches bugs.
- **Signal:** binary reward from a validation program over database state; rejection fine-tuning keeps only passing runs.



Our redraw (not the paper’s figure): clone, verify, then train.

Key results: online success rate (%)



Tables 4 and 5, redrawn; no std or CI given.

Qwen3-4B: 7.88 to 13.94 (WA-Lite), 13.18 to 20.45 (M2W-O)

SO WHAT → *LLaMA-3.2-3B on M2W-Online: ADP 24.09, VeriEnv 24.55; the authors attribute ADP’s gain there to dataset overlap with its training data. On Qwen3-4B, ADP gives 6.06 and 11.36. Proprietary references on WA-Lite: GPT-4o 31.52, GPT-4o-mini 23.64 (numbers reported by another paper).*

Human check of recreated sites (%)

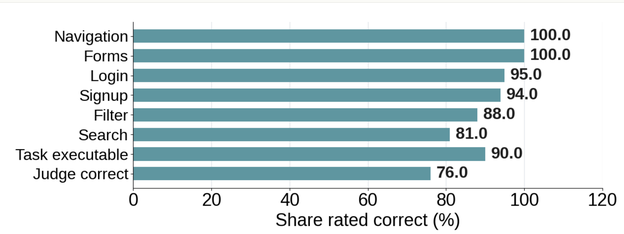


Table 3, redrawn: 4 annotators; first six rows are functional checks (average 90%).

SO WHAT → *Judge correctness is the lowest (76%); the most common judge errors came from database resets that did not preserve the random seeds. The authors report that re-running the SDK validation programs detects and resolves such errors. Inter-annotator agreement is substantial (mean κ 0.61). Annotators rated visual quality 4.7 on a 5-point scale. Environment checks cover 4 annotators with double annotation per subset.*

Headline Numbers

13.94%
WebArena-Lite, Qwen3-4B
base 7.88%

- **20.45%** M2W-Online, Qwen3-4B
- **12.73%** WA-Lite, LLaMA-3.2-3B
- **+6.06** WA-Lite Δ, Qwen3-4B

The paper’s own controls

CONTROL	VERIENV	COMPARED WITH
M2W-O, LLaMA-3.2-3B	24.55	ADP 24.09 (overlap)
WA-Lite, Qwen3-4B	13.94	GPT-4o 31.52 (reported)
M2W-O, Qwen3-4B	20.45	Browser-Use-GPT-4o 30.00
Seeds / variance	not reported	no std or CI given

Takeaway

Clone sites, verify by database state, fine-tune on passing runs: online gains over base, low absolute rates, no ablation.

149 sites, 7,400 tasks | 83.5 min, $3.6 per site | judge correct 76%

Scan to Read

Paper | Code